Supporting Information

Computational Generation Time as a Proxy for Evolutionary Accessibility in Lignin: A Two-Experiment Falsification Framework

Combariza *et al.*, 2026

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S1. Kolmogorov Complexity and the t_{gen} Proxy

S1.1. Kolmogorov Complexity: Definition and Properties

Definition S1.1 — Kolmogorov Complexity

Let U be a universal Turing machine (UTM), x a binary string, and p a program (binary string). The **Kolmogorov complexity** of x is:

$$K_U(x) \;=\; \min_{p\,:\,U(p)=x} \ell(p)$$

where $\ell(p)$ is the length of program p in bits.

$K(x)$ is the length of the *shortest computer program* that generates x and halts. It measures the minimum information content intrinsic to x: a string with internal regularities (repetitions, patterns) has low K; a truly random string has $K(x) \approx \ell(x)$.

Uncomputability. $K(x)$ is uncomputable as a consequence of the Halting Problem. No algorithm can compute $K(x)$ exactly for arbitrary x. Practical applications therefore rely on approximations: lossless compression (upper bound), the Coding Theorem Method (CTM; Delahaye & Zenil 2012) applicable to short strings, or the Block Decomposition Method (BDM) for longer objects.

Invariance Theorem. For any two universal Turing machines U and V:

$$|K_U(x) - K_V(x)| \leq c_{UV}$$

where c_{UV} is a machine-dependent constant independent of x. This makes $K(x)$ an intrinsic property of the string, not an artifact of the computational model.

S1.2. Algorithmic Probability and the Coding Theorem

Definition S1.2 — Algorithmic Probability (Solomonoff, 1964)

$$P_U(x) \;=\; \sum_{\substack{p\,:\,U(p)=x \\ \text{halts}}} 2^{-\ell(p)}$$

The probability that a randomly drawn binary program causes U to output x.

Theorem S1.1 — Coding Theorem (Levin, 1974)

$$\log_2 P_U(x) \;\geq\; -K_U(x) \;-\; O(1) \quad \Leftrightarrow \quad \boxed{P_U(x) \;\geq\; 2^{-K_U(x)-c}}$$

Reading the equation. Lower $K(x)$ (shorter description) implies higher $P(x)$ (greater probability of arising by random process). The bound is exponential: reducing $K(x)$ by n bits multiplies $P(x)$ by at least 2^n.

Evolutionary implication (Johnston *et al.* PNAS 2022): phenotypes with lower K are exponentially overrepresented in biological genotype–phenotype maps. This "simplicity bias" shapes evolutionary trajectories independently of natural selection ("arrival of the frequent", Wagner 2014).

S1.3. The t_{gen} Proxy: Formal Relationship

The LGS algorithm is a rejection sampler. Let $P_{\text{LGS}}(S \mid \theta)$ be the probability that one draw from the LGS prior produces a valid structure S given parameters $\theta = (f_{\beta\text{-O-4}}, r_{\text{S/G}}, \text{DP})$. Then the expected number of iterations before acceptance is:

$$R(S;\theta) \;=\; \frac{1}{P_{\text{LGS}}(S \mid \theta)}$$

and wall-clock time is:

$$t_{\text{gen}}(S;\theta) \;\approx\; t_{\text{step}} \cdot R(S;\theta) + t_{\text{overhead}} \;\propto\; \frac{1}{P_{\text{LGS}}(S \mid \theta)}$$

If the LGS prior approximates the chemical probability distribution of lignin polymerization:

$$P_{\text{LGS}}(S \mid \theta) \;\approx\; 2^{-\tilde{K}(S|\theta)}$$

then $t_{\text{gen}} \propto 2^{\tilde{K}(S|\theta)}$ — an exponential relationship between generation time and conditional descriptional complexity.

S1.4. Comparison: $K(x)$ vs. t_{gen}

Property	$K(x)$	t_{gen}
Computability	Uncomputable	Directly measurable
Universality	UTM-invariant (up to additive constant)	LGS-specific; not universal
Formal connection to $P(x)$	Coding Theorem: $P(x) \geq 2^{-K(x)-c}$	Rejection rate: $t_{\text{gen}} \propto P_{\text{LGS}}^{-1}$
What it measures	Intrinsic complexity of x	Conditional complexity given LGS prior θ
String length	Applicable to all binary strings	Applied here: SMILES of DP$\in [5, 10]$
Algorithm invariance	Yes (Invariance Theorem)	No — t_{gen} is LGS-specific
Evolutionary claim	$K(x) \downarrow \Rightarrow P(x) \uparrow$ (Coding Theorem)	$t_{\text{gen}} \downarrow \Rightarrow P_{\text{LGS}} \uparrow \Rightarrow$ *possibly* favored
Validation in biology	Johnston *et al.* PNAS 2022 (proteins, RNA)	This work (lignin, first application)

Key point

The proxy is empirically justified, not formally derived. We do not claim $t_{\text{gen}} = K(x)$ or that t_{gen} satisfies the Invariance Theorem. We claim that t_{gen} is a computable, reproducible measure that captures domain-specific conditional complexity, and test empirically whether this quantity predicts evolutionary trends in lignin.

S2. Complete Statistical Analysis

S2.1. Experiment 1: Full ANOVA Table

Table S1: Two-way ANOVA for Experiment 1 (t_{gen} as dependent variable; $n = 270$, residual $df = 261$). SS = sum of squares; MS = mean square; η_p^2 = partial eta-squared.

Source	**SS**	*df*	**MS**	*F*	η_p^2
Plant type (group)	8406.7	2	4203.4	1342.8	0.911
DP	66383.1	2	33191.6	10603.2	0.988
Plant type × DP	26503.0	4	6625.7	2116.6	0.970
Residual	817.0	261	3.13		

All $p < 0.001$ (plant type: $p = 4.2 \times 10^{-138}$; DP: $p = 1.2 \times 10^{-250}$; interaction: $p = 1.6 \times 10^{-197}$)

S2.2. Experiment 1: Tukey HSD Post-Hoc Tests

Table S2: Tukey HSD pairwise comparisons for Experiment 1, factor: plant type (pooled over DP levels). Hedges' g used as effect size.

A	**B**	$\bar{t}_A$	$\bar{t}_B$	Δ	g	p_{Tukey}
GR	HW	19.64	30.00	−10.4	−0.473	< 0.001
HW	SW	30.00	17.11	+12.9	+0.615	< 0.001
GR	SW	19.64	17.11	+2.5	+0.213	0.636 (ns)

Table S3: Tukey HSD pairwise comparisons for Experiment 1, factor: DP (pooled over plant types).

A	**B**	$\bar{t}_A$	$\bar{t}_B$	Δ	g	p_{Tukey}
DP=10	DP=5	42.34	4.07	+38.3	+2.760	< 0.001
DP=10	DP=8	42.34	20.34	+22.0	+1.547	< 0.001
DP=5	DP=8	4.07	20.34	−16.3	−5.023	< 0.001

S2.3. Experiment 1: Normality and Homoscedasticity

S2.4. Experiment 2: Full ANOVA Table

S2.5. Experiment 2: Per-Cell Descriptive Statistics

S2.6. Experiment 2: Tukey HSD Post-Hoc Tests

S2.7. Non-Parametric Robustness Check

S2.8. Post-Hoc Power Analysis

Table S4: Shapiro–Wilk normality test per cell for Experiment 1 ($n = 30$ per cell). CV = coefficient of variation (%). * indicates $p < 0.05$ (mild non-normality); all ANOVA results confirmed by non-parametric Kruskal–Wallis tests.

Group	DP	W	p	CV (%)
SW	5	0.944	0.118	6.3
SW	8	0.910	0.015*	8.2
SW	10	0.954	0.222	5.1
HW	5	0.951	0.174	6.4
HW	8	0.975	0.693	11.4
HW	10	0.977	0.750	4.8
GR	5	0.918	0.024*	5.4
GR	8	0.906	0.012*	8.4
GR	10	0.953	0.206	6.1

Table S5: Two-way ANOVA for Experiment 2 (t_{gen} as dependent variable; $n = 239$, residual $df = 230$).

Source	SS	df	MS	F	η_p^2
$f_{\beta\text{-O-4}}$	1,540,810	2	770,405	284.6	0.712
S/G ratio	351,765	2	175,883	65.0	0.361
$f_{\beta\text{-O-4}}\times$S/G	210,995	4	52,749	19.5	0.253
Residual	622,613	230	2,707		

All $p < 0.001$ ($f_{\beta\text{-O-4}}$: $p = 6.2 \times 10^{-63}$; S/G: $p = 4.3 \times 10^{-23}$; interaction: $p = 8.0 \times 10^{-14}$)

S3. LGS Generation Machinery and the HW Complexity Peak

S3.1. The Four-Stage Generation Pipeline

The LGS algorithm (Eswaran *et al.* 2022) operates in four stages documented in their Methods section:

Stage 1 — Monomer sequencing (Heap's algorithm). All unique permutations of G and S monomers are generated for the given S/G ratio and DP. The number of unique sequences is:

$$N_{\text{seq}} = \binom{n}{k}$$

where $n = \text{DP}$ and k = number of S units, determined by $k = \lfloor n \cdot r_{\text{S/G}}/(1 + r_{\text{S/G}}) \rceil$.

Stage 2 — Conditional linkage generation. For each monomer sequence, bond types are sampled conditionally on local monolignol identity. Table S11 reproduces the coupling rules from Table 3 of Eswaran *et al.* (2022).

Stage 3 — Rejection sampling. Candidate structures are accepted only if they satisfy the `bondconfig` frequency targets. Structures that fail are discarded and re-drawn. This is the stage where t_{gen} varies.

Stage 4 — SMILES output. Accepted structures are converted to canonical SMILES via the Chemistry Development Kit (CDK).

Table S6: Per-cell descriptive statistics for Experiment 2. Mean and SD of t_{gen} (seconds). CV = coefficient of variation (%). n = number of successful runs (of 30 planned).

$f_{\beta\text{-O-4}}$	S/G	n	Mean (s)	SD (s)	CV (%)
0.54 (SW)	0.0	27	100.83	42.52	42.2
0.54 (SW)	0.8	28	116.45	18.48	15.9
0.54 (SW)	1.5	27	108.46	27.53	25.4
0.69 (HW)	0.0	24	185.37	35.54	19.2
0.69 (HW)	0.8	28	374.30	82.79	22.1
0.69 (HW)	1.5	27	288.86	90.60	31.4
0.77 (GR)	0.0	27	72.46	17.80	24.6
0.77 (GR)	0.8	25	152.32	24.80	16.3
0.77 (GR)	1.5	26	140.60	62.01	44.1

Table S7: Tukey HSD and Cohen's d for Experiment 2, factor: $f_{\beta\text{-O-4}}$ (pooled over S/G levels).

A	**B**	$\bar{t}_A$	$\bar{t}_B$	Δ (s)	d	p_{Tukey}
HW (0.69)	SW (0.54)	287.7	108.7	+179.0	+2.28	< 0.001
HW (0.69)	GR (0.77)	287.7	120.8	+166.9	+1.97	< 0.001
GR (0.77)	SW (0.54)	120.8	108.7	+12.1	+0.28	0.528 (ns)

S3.2. Why the HW Complexity Peak Is Not an Artifact

The combinatorial argument

At DP = 10, HW ($r_{\text{S/G}} = 1.5$) and GR ($r_{\text{S/G}} = 0.8$) have *identical* sequence spaces:

$$N_{\text{seq}}(\text{HW}) = \binom{10}{6} = 210 \qquad N_{\text{seq}}(\text{GR}) = \binom{10}{4} = 210$$

Yet HW generates 2.3× slower (283 s vs. 122 s). The difference cannot be attributed to enumeration size. It arises from the bond constraint: HW requires 31% C–C bonds in its `bondconfig`, but C–C bonds are restricted to G-unit positions (Table S11). With 6 S units and 4 G units in the HW pool, the algorithm must repeatedly find G-unit positions for C–C bonds while placing S units elsewhere — a combinatorially constrained search. GR requires only 23% C–C bonds and has 77% β-O-4 (accessible to both G and S), which dissolves the constraint.

This argument demonstrates that the HW complexity peak is a consequence of the *chemically correct* encoding of monolignol bonding restrictions in LGS (Eswaran *et al.* 2022, Table 3), not of any hardcoded optimization or tuning of the algorithm for specific $f_{\beta\text{-O-4}}$ values.

Table S8: Tukey HSD and Cohen's d for Experiment 2, factor: S/G ratio (pooled over $f_{\beta\text{-O-4}}$ levels).

A	**B**	$\bar{t}_A$	$\bar{t}_B$	Δ (s)	d	p_{Tukey}
S/G=0.8	S/G=0.0	216.7	117.0	+99.6	+1.00	< 0.001
S/G=1.5	S/G=0.0	179.8	117.0	+62.8	+0.75	< 0.001
S/G=0.8	S/G=1.5	216.7	179.8	+36.9	+0.32	0.053 (ns)

Table S9: Kruskal–Wallis tests for Experiment 2, confirming ANOVA results under no distributional assumptions.

Factor	H	df	p
$f_{\beta\text{-O-4}}$	135.6	2	3.6×10^{-30}
S/G ratio	42.4	2	6.3×10^{-10}

S4. Parameter Grounding and Experimental Configurations

S4.1. Native MWL Literature Ranges

S4.2. Normalized LGS Configurations for All 18 Cells

Table S10: Post-hoc statistical power for key comparisons (two-sample t-test, $\alpha = 0.05$, $n = 30$ per group). Power computed from observed effect sizes.

Comparison	d	n	**Power**
Exp. 2: HW vs. SW	2.28	30	> 0.9999
Exp. 2: HW vs. GR	1.97	30	> 0.9999
Exp. 2: SW vs. GR	0.28	30	0.191
Exp. 1: SW vs. HW at DP=8	1.50	30	> 0.9999

Note: The study was deliberately underpowered to detect the small SW vs. GR effect ($d = 0.28$); $n \approx 200$ would be required for 80% power at this effect size.

Table S11: Monomer coupling patterns reproduced from Eswaran *et al.* (2022), Table 3. ✓ = bond type possible; × = bond type forbidden. S units lack the free C-5 position of G units, preventing β-5 and 5-5 C–C bonds.

		Linear			**Branched**	
M1	**M2**	β-O-4	β-β	β-5	4-O-5	5-5
G	G	✓	✓	✓	✓	✓
S	S	✓	✓	×	×	×
S	G	✓	✓	✓	×	×
G	S	✓	✓	×	✓	×

S5. Execution Environment and Reproducibility

S5.1. Hardware and Software

S5.2. Parallel Execution Sandboxing

LGS reads from a fixed `resources/project_config.yaml` path and writes to fixed `output/` subdirectories relative to the working directory. To enable parallel execution across 8 workers without file-system conflicts, 8 isolated working directories (`worker_0` through `worker_7`) were created, each containing:

- A symbolic link to the shared `lgs_gen.jar`
- Independent `resources/` subdirectory with the cell-specific `project_config.yaml`
- Independent `output/` subdirectory for LGS output

Jobs were distributed across workers via `ThreadPoolExecutor(max_workers=8)` with randomized task order to prevent systematic load imbalance. Output files were deleted immediately after t_{gen} was recorded to manage disk usage.

S5.3. Data and Code Availability

All configuration files, analysis notebooks, and raw results are publicly available at:

`https://github.com/In-Silico-RG/LSG_Molec_Evol`

Repository contents:

- `configs/experiment1/` — 9 YAML files (SW/HW/GR × DP5/8/10)
- `configs/experiment2/` — 9 YAML files ($f_{\beta\text{-O-4}}$× S/G orthogonal)

Table S12: Combinatorial analysis of sequence spaces and bond constraints at DP = 10 for the three plant types. "C–C req." = fraction of non-β-O-4 bonds in `bondconfig` (approx. $1 - f_{\beta\text{-O-4}}$ accounting for all C–C and mixed bond types).

Config	$\binom{10}{k}$	**C–C req.**	$\bar{t}_{\text{gen}}$ (s)	**Bottleneck**
SW (S/G= 0)	$\binom{10}{0} = 1$	46%	109	G-only: no conditional constraint
HW (S/G= 1.5)	$\binom{10}{6} = 210$	31%	283	C–C bonds require G positions in G+S pool
GR (S/G= 0.8)	$\binom{10}{4} = 210$	23%	122	High β-O-4 dissolves G-position constraint

Table S13: Native milled wood lignin (MWL) interunit linkage frequencies (% of total bonds) used to define parameter bounds for LGS configuration. Values represent experimental data from primary MWL studies. Central values (midpoint of range) were used in all experiments and normalized to sum to 100 for the `bondconfig` weight format.

Plant type	β-O-4	β-5	β-β	4-O-5	5-5
Softwood (SW)	43–50	9–12	2–4	4–8	18–25
Hardwood (HW)	50–65	4–8	3–8	5–8	5–12
Grass (GR)	65–80	3–7	3–6	5–10	3–8

Sources: Eswaran *et al.* (2022); Obrzut *et al.* (2023); Lourenço & Pereira (2018); Chakar & Ragauskas (2004).

- `notebooks/05-Parallel_Experiment.ipynb` — main execution notebook (8-worker parallel, auto-generates YAMLs, no external files needed beyond `lgs_gen.jar`)
- `results/evolutionary_experiment_results.csv` — raw data, Experiment 1 (270 rows)
- `results/exp2_parallel_results.csv` — raw data, Experiment 2 (244 rows, 239 successful)

Table S14: Complete `bondconfig` parameters (normalized to sum to 100) for all experimental configurations. Experiment 1 configurations are identified by plant type and DP; Experiment 2 configurations by $f_{\beta\text{-O-4}}$ source and S/G level. All Experiment 2 runs at DP = 10.

Exp.	**Config**	**DP**	f_β	S/G	BO4	B5	BB	55
1	SW_DP5	5	0.54	0.0	54	11	3	25
1	SW_DP8	8	0.54	0.0	54	11	3	25
1	SW_DP10	10	0.54	0.0	54	11	3	25
1	HW_DP5	5	0.69	1.5	69	7	7	10
1	HW_DP8	8	0.69	1.5	69	7	7	10
1	HW_DP10	10	0.69	1.5	69	7	7	10
1	GR_DP5	5	0.77	0.8	77	5	4	6
1	GR_DP8	8	0.77	0.8	77	5	4	6
1	GR_DP10	10	0.77	0.8	77	5	4	6
2	SW_fBO4_sg0	10	0.54	0.0	54	11	3	25
2	SW_fBO4_sg08	10	0.54	0.8	54	11	3	25
2	SW_fBO4_sg15	10	0.54	1.5	54	11	3	25
2	HW_fBO4_sg0	10	0.69	0.0	69	7	7	10
2	HW_fBO4_sg08	10	0.69	0.8	69	7	7	10
2	HW_fBO4_sg15	10	0.69	1.5	69	7	7	10
2	GR_fBO4_sg0	10	0.77	0.0	77	5	4	6
2	GR_fBO4_sg08	10	0.77	0.8	77	5	4	6
2	GR_fBO4_sg15	10	0.77	1.5	77	5	4	6

Table S15: Execution environment for all LGS runs.

Component	**Version / Specification**
Operating system	Ubuntu 24.04 LTS
Java runtime	OpenJDK 17 (LGS is a Java process)
Python	3.13
Statistical analysis	`scipy` 1.13, `pingouin` 0.5, `pandas` 2.2
Parallel workers	8 (`ThreadPoolExecutor`)
Timing measurement	`time.time()` bracketing `subprocess.run()`
LGS version	`lgs_gen.jar` (Eswaran *et al.* 2022)
LGS source code	`github.com/sudhacheran/lignin-structure-generator`